

# The cycle in a syllable:

## Phases and impenetrability in Dinka morphology\*

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### *Abstract*

This paper argues for a cross-modular theory of cyclicity rooted in phase theory (Chomsky 2001; Marvin 2002; Newell 2008 a.o.), in which the application of the syntactic, morphological, and phonological grammar is interleaved. I present a detailed investigation of the Nilotic language Dinka (South Sudan), a language with extensive effects of cyclicity in its syntax and a rich system of nonconcatenative morphology. I show that several asymmetries in how nonconcatenative morphology is expressed on the verb, including allomorphy, the availability of affixation, and shifts in voice quality, align with a clause-internal domain diagnosed by successive-cyclic long-distance movement. As a result, Dinka provides a clear argument that the domains of cyclic computation of phonology, morphology, and syntax are parallel, as predicted by a phase-theoretic view of cyclicity. In addition, the nonconcatenative morphology of Dinka demonstrates a key property of an interleaved system: morphosyntactic information is rendered opaque at the end of each cycle of interpretation, but phonological content remains accessible in higher domains.

## 1 Introduction

A key question in linguistic theory is the extent to which the subcomponents of grammar make use of the same principles and operations. A cognitive system with fewer primitives is more parsimonious and evolutionarily plausible, and there are many surface similarities across modules, such as the use of binary hierarchical structures or effects like doubling and deletion. At the same time, the modules of grammar operate on different inputs and outputs and so there must naturally be differences in computation also.

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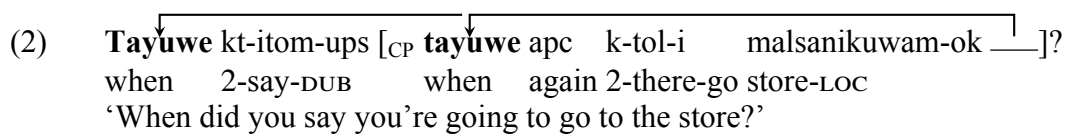
\*My thanks to David Adger, Athulya Aravind, Hagit Borer, Adam Chong, Paula Fenger, Laura Kalin, David Pesetsky, Nick Rolle, Linnaea Stockall, and Michelle Yuan for comments and discussion, as well as audiences of talks at the Verbal Domains workshop in Newcastle in 2023 and NELS55 at Yale in 2024. Abbreviations used for Dinka: AP = antipassive, APPL = applicative, CF = centrifugal, CP = centripetal, D = declarative, FUT = future, GEN = genitive, NEG = negation, NF = non-finite, PASS = passive, OBLV = Oblique Voice, OV = Object Voice, PRF = perfect, SV = Subject Voice. Email: c.vanurk@qmul.ac.uk

One possible cross-modular property is *cyclicity*, the observation that linguistic structure is built in stages. Cyclicity has been taken to be a core property of morphology and phonology since Chomsky and Halle (1968). Cyclicity is evident, for instance, in the inheritance of phonological processes in derived forms, such as in the well-known contrast in syllabification of root-final sonorants in *-ing* nominalizations in (1a–b).

- (1) a. [N *twinkl-ing*] [twɪŋ.klɪŋ] ‘instant’  
 b. [N [V *twinkl*]-*ing*] [twɪŋ.kl̩.ɪŋ] ‘an event of twinkling’  
 (Chomsky and Halle 1968:86)

This contrast can be understood if syllabification occurs on the cycle associated with the verb inside of the nominalization (Marvin 2002:31–41), so that the complex nominalization in (1b) must inherit the syllabic structure of the verb in isolation.

Cyclicity has played an important role in syntactic theory too, since the early days of transformational grammar (Chomsky 1965, 1973). Many languages offer evidence, for instance, that long-distance movement rules like *wh*-movement proceed successive-cyclically, and occur in each clause on the path of movement. In Passamaquoddy, such intermediate steps are evident in the overapplication of *wh*-movement, resulting in *wh*-copying (2).

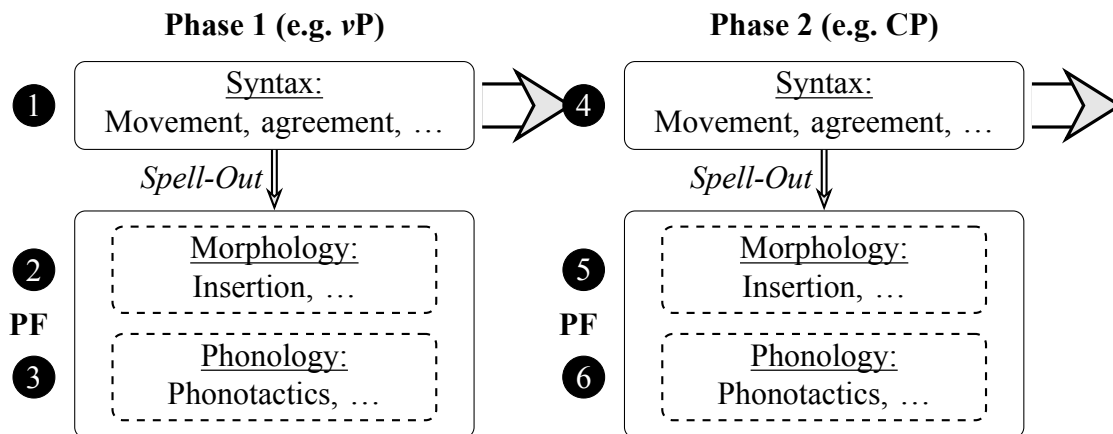
- (2)   
 when 2-say-DUB when again 2-there-go store-LOC  
 ‘When did you say you’re going to go to the store?’  
 (Passamaquoddy; Bruening 2006:26)

This observation follows if each clause is a syntactic cycle in which *wh*-movement is triggered.

In contemporary minimalist work, an influential cross-modular approach to cyclicity comes from phase theory (Chomsky 2001 et seq.). In this theory, syntactic structures are built in *phases*. Upon completion of a phase, the content of the phase is mapped onto morphological and phonological structures at Phonological Form (PF), through an operation of Spell-Out. This model results in the *interleaving* of syntactic, morphological, and phonological operations, since a phase is interpreted by each module before the next phase is built, as visualized in (3).<sup>1</sup>

<sup>1</sup>I adopt an architecture with separate morphological and phonological modules, in which morphological interpretation precedes the application of the phonological grammar (Halle and Marantz 1993; Arregi and Nevins

(3) *Cyclic interleaving in phase theory:*



Because this model is syntax-first, cycles in syntax end up creating cycles of morphology and phonology, providing a unified explanation for the existence of cycles across the grammar.

Within phase theory, cyclic rule application in syntax is typically enforced by means of an accessibility condition, which limits the scope of syntactic rules to the current phase. Chomsky (2001) proposes the Phase Impenetrability Condition (PIC), which renders syntactic structure that has undergone Spell-Out inaccessible.

(4) **Phase Impenetrability Condition:**

After Spell-Out of a phase, material within it becomes *inaccessible* to later operations, except for the phase edge.

Phase impenetrability limits the ability of syntactic heads to access structure in another phase. To allow for long-distance dependencies, the PIC builds in an “escape hatch”, the phase edge, which derives the need for successive-cyclic derivations, as evident in Passamaquoddy.

A cross-modular theory of cyclicity faces several challenges. First, evidence for cyclicity in morphophonological processes comes from cases in which a rule is limited to a subword domain, as in (1b). But, although there is a broad consensus in syntactic work on cyclicity that the clause constitutes a locality domain (*cf.* Den Dikken 2009), the existence of a syntactic clause-internal phase remains subject to debate (e.g. Grano and Lasnik 2018; Keine 2020; Keine and Zeijlstra 2025), and it has been questioned whether such a domain boundary would line up

2012), though the conclusions argued for here are compatible with alternative conceptions of the relationship between these also.

with subword domains in morphophonological computation (Cheng and Downing 2016). A second issue relates to whether an accessibility condition like phase impenetrability explains opacity in morphological and phonological processes also. There are morphophonological effects that fail to apply across apparent phase boundaries (Marvin 2002; Newell 2008; Embick 2010; Samuels 2012; Fenger 2020; Sande et al. 2020). But, at the same time, it is well-known that there are morphophonological rules that do seem to be able to alter the content of previous cycles (Lowenstamm 2010; Embick 2013).

Significant questions remain then about the prospects of a cross-modular theory of cyclicity rooted in phases. One difficulty in resolving these questions reliable diagnostics for syntactic and morphophonological cycles for the same domain within one language can be elusive (d’Alessandro and Scheer 2015). This paper demonstrates that the Western Nilotic language Dinka (South Sudan) provides a useful testing ground for theories of cyclicity across modules. Dinka (Nilotic, South Sudan) shows extensive effects of cyclicity in its syntax (Van Urk 2015, 2018; Van Urk and Richards 2015), and these effects have played an important role in the debate around the existence of a clause-internal phase (see Keine and Zeijlstra 2025). In addition, like many Western Nilotic languages, it has a rich system of nonconcatenative morphology within a monosyllabic root (Andersen 1993 et seq), as shown in Table 1.

**Table 1. Verbal paradigm of *mîit* ‘pull’ in Agar Dinka (Andersen 2017:12).**

|     | Transitive | CF     | CP   | APPL   | AP     |
|-----|------------|--------|------|--------|--------|
| NF  | mîit       | mǰêet  | mǰet | mǰet   | mǰit   |
| 1sg | mǰet       | mǰet   | mǰet | mǰet   | mǰet   |
| 2sg | mîit       | mǰet-é | mǰit | mǰet-é | mǰet-é |
| 3sg | mîit       | mîit   | mǰit | mîit   | mǰit   |
| ... | ...        | ...    | ...  | ...    | ...    |

The pervasiveness of nonconcatenative marking in this system means that many morphemes are realized in exactly the same position, the monosyllabic root, and via the same morphophonological changes, two recurring processes of vowel lowering and vowel fronting. As a result, if

there are abstract domain boundaries in morphology and phonology, clear differences should emerge in how morphemes realizing higher and lower heads in the extended verbal projection are able to interact with the root.

The first conclusion that emerges from the comparison of phonology, morphology, and syntax in Dinka is that the cycles of these modules are *parallel*, as predicted by phase theory. Syntactic diagnostics for a clause-internal *vP* phase, as identified in Van Urk and Richards (2015) and Van Urk (2015, 2018), line up with a well-known distinction between inner and outer morphology in Dinka, such that all *inner* morphemes are generated in the *vP* phase. In addition, the root plus inner morphology constitutes a morphophonological cycle, over which the monosyllabic template is enforced and across which allomorphy based on morphosyntactic features is not possible. I derive both observations from phasal locality, developing a model in which phases undergo phonological interpretation and prosodification cyclically and in which only morphemes spelled out in the same cycle are active for allomorphic relations (Embick 2010; Bobaljik 2012; Moskal 2015). As a result, Dinka provides novel evidence for the idea that cycles across modules correspond, as in phase theory, and for the notion of a clause-internal phase (Chomsky 2001, Van Urk and Richards 2015; contra Keine and Zeijlstra 2025).

Within this system, the hypothesis that phase impenetrability places constraints on what material is accessible to morphophonological rules can also be tested directly. In Dinka, phasal locality restricts the application of syntactic movement as well as allomorphy. I demonstrate that some phonological properties of the root, such as voice quality, become fixed in the *vP* phase also and can only be altered by inner morphemes. At the same time, however, inner and outer morphemes trigger the *same* processes of nonconcatenative morphology, altering vowel quality, length, and tone. Dinka then offers evidence *against* an accessibility condition like phase impenetrability for phonological rules. Phonological material remains accessible to outer morphemes. I derive the asymmetry between morphological and syntactic rules, on the one hand, and phonological rules, on the other, from the idea that Spell-Out is a *replacive* operation (see also Adger 2007, cf. Bobaljik 2000), overwriting the syntactic features of a node with its phonological content. As a result, rules that act on syntactic features are subject to phase impenetrability, but phonological structure, the output of PF interpretation, remains accessible.

I argue instead that cases of opacity in phonological computation arise from phase theory in a different manner. I propose a family of phase faithfulness constraints in an Optimality-theoretic grammar (Michaels 2013; Šurkalović 2013; McPherson and Heath 2016), which can enforce faithfulness to the output of a previous phase. Such constraints will not restrict the expression of morphemes in inner domains, so that such morphemes may affect greater changes.

The paper is structured as follows. I first review syntactic evidence for a  $\nu$ P phase in Dinka, based on the distribution of verb phrase V2 and pronoun copying (Van Urk 2015, 2018; Van Urk and Richards 2015), and demonstrate that such a boundary lies roughly between the thematic domain and the domain of TAM morphology. Section 3 introduces the nonconcatenative morphology of Dinka verbs and shows that, within such morphemes, there is a key distinction between outer and inner morphemes (Andersen 1992, 1993 et seq.), which aligns with the  $\nu$ P phase boundary. Section 4 offers evidence that, in accordance with the predictions of phase theory, the verb and inner morphemes are a locality domain for allomorphy and prosodification, so that they map obligatorily to a monosyllabic word. Finally, in section 5, I show that phonological rules may treat the  $\nu$ P phase as a cyclic domain, for shifts in voice quality, but that inner and outer morphemes are equally able to alter the vowel quality, length, and tone of the verb. These observations show that there is no strict phase impenetrability in phonology, unlike in morphology and syntax, a consequence of cyclic interleaving.

## **2 Successive cyclicity in Dinka and the $\nu$ P phase**

This section introduces the syntactic evidence for a clause-internal cycle in Dinka (Van Urk 2015, 2018; Van Urk and Richards 2015). In Dinka, the left edge of the verb phrase hosts movement of an object, if possible, resulting in a verb-second (V2) effect. As Van Urk (2015, 2018) and Van Urk and Richards (2015) observe, this object position is obligatorily targeted for intermediate successive-cyclic movement in long-distance dependencies. First, the phrase undergoing long-distance movement satisfies the V2 property of each intervening verb phrase, so that object positions that must otherwise be filled are empty in the context of extraction (5).

- (5) Yè **ŋó** cǐi Áyèn [<sub>VP</sub>        cǝk [cǝm Ból       ]]?  
 be what PRF.OV Ayen.GEN make.NF eat.OV Bol.GEN  
 ‘What has Ayen made Bol eat?’

Second, in long-distance dependencies involving plurals, every object position on the path of movement is occupied by a 3PL pronoun *kêek*, argued by Van Urk (2018) to be the realization of an intermediate copy.

- (6) Yè **kôc-kò** yǐi Ból [<sub>VP</sub> **kêek** luêeel ɛ cǐi Áyèn [<sub>VP</sub> **kêek** tǐi]]?  
 be.3SG people-which be.OV Bol.GEN 3PL say.NF C PRF.OV Ayen.GEN 3PL see.NF  
 ‘Which people is Bol saying that Ayen has seen?’

This section reviews these effects and concludes, contra Keine and Zeijlstra (2025), that they derive from the presence of a *vP* phase in Dinka, demarcated by an object position before the in-situ verb.

## 2.1 V2 in the verb phrase

In order to investigate the hypothesis that locality domains in syntax, morphology, and phonology are parallel, I first describe the syntactic evidence for a clause-internal domain in Dinka (Van Urk 2015, 2018; Van Urk and Richards 2015). In this section, I demonstrate that there is an object position before in-situ verbs, which is targeted by obligatory intermediate movement.

Dinka is a Western Nilotic language, spoken in South Sudan and in a diaspora community primarily located in the US and Australia. Van Urk (2015, 2018) and Van Urk and Richards (2015) draw on data from the Dinka Bor variety. All the syntactic data I present here is drawn from Bor, though I focus mainly on the Agar variety described by Andersen (1992, 1993) when discussing the morphophonology. As far as I can determine, the key syntactic, morphological, and phonological generalizations hold across varieties.

Dinka has a verb-second effect at the level of the clause, so that the highest verb/auxiliary obligatorily moves to second position. In neutral declaratives, the second position verb or auxiliary follows the subject (7a), but, as in familiar V2 languages, the highest verb or auxiliary inverts with the subject after a topicalized phrase (7b).

(7) *Clause-level verb-second in Dinka:*

- a. **Ayén** à-càm cuḡin nè pǎal.  
Ayen 3s-eat food P knife  
'Ayen is eating food with a knife.'
- b. **Cuḡin** à-céem Áyèn nè pǎal.  
food 3s-eat.OV Ayen.GEN P knife  
'Food, Ayen is eating with a knife.'

The main focus of this paper, though, is on a similar V2 effect that obtains in the verb phrase, involving object movement. Dinka has a clause-medial position that is obligatory occupied by an object, if possible. This position is followed by any verbs or auxiliaries that have not raised into the left periphery, thus creating another V2 configuration. In (8a–b), the transitive object *mìir* appears in this clause-medial position, which I will argue demarcates the left edge of the verb phrase. Because the future auxiliary *bé* is in the clause-level V2 position, the verb *tḡin* is in situ and must follow the object.

(8) *V2 in the verb phrase with in situ verb:*

- a. Yḡin bé [<sub>VP</sub> **mìir** tḡin]  
2SG FUT giraffe see.NF  
'You will see a giraffe.'
- b. \*Yḡin bé [<sub>VP</sub> \_\_\_\_ tḡin **mìir**].  
2SG FUT see.NF giraffe  
'You will see a giraffe.'

The same effect obtains in ditransitives. One object must sit in the clause-medial position and so will precede an in situ verb, if present, while the other must follow the verb (9a–b).

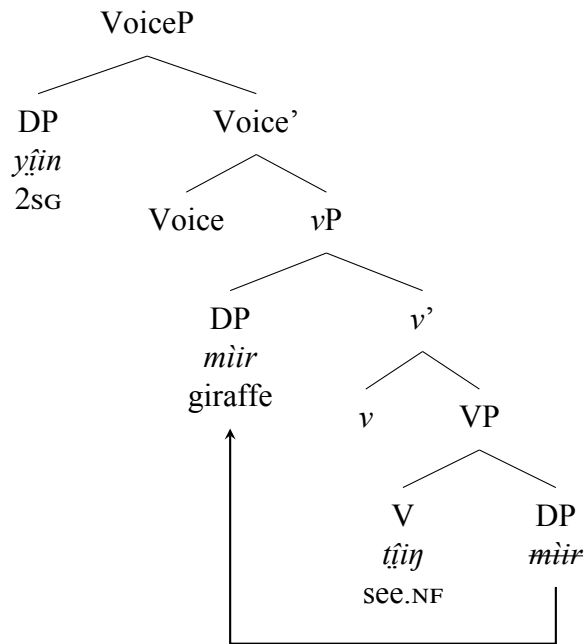
(9) *One object before the verb in ditransitives:*

- a. Yḡin cé [<sub>VP</sub> **Bòl** gâam cáa akòl].  
2SG PRF Bol give.NF milk day  
'You have given Bol milk in the afternoon.'
- b. Yḡin cé [<sub>VP</sub> **cáa** gâam Bòl akòl].  
2SG PRF milk give.NF Bol day  
'You have given Bol milk in the afternoon.'

Van Urk (2015, 2018) and Van Urk and Richards (2015) explain this pattern by proposing that Dinka has obligatory object shift to a clause-medial specifier, which I identify with a

specifier of a  $\nu$ P below VoiceP (10).

(10) *Structure of Dinka  $\nu$ P:*



The verb phrase is otherwise head-initial, and, as a result, an in situ verb will end up immediately following the moved object.

I refer to the phrase hosting object movement as  $\nu$ P. The  $\nu$ P has been associated with a range of functions in minimalist work: introducing the external argument, hosting object movement and licensing objects, categorizing roots, and introducing a phase boundary. The phrase I call  $\nu$ P in Dinka clearly fulfils at least two of these functions: it hosts object shift and introduces a phase boundary. I associate the introduction of the external argument with a higher head Voice, explaining why the subject does not count for the verb phrase V2, though an account in which object shift targets an inner specifier of Voice is also conceivable (with VoiceP the phase). The phrase that hosts object shift clearly does not have a categorizing function, however, as we will see evidence in section 2.3 for functional structure below the object position.

An object movement analysis receives further support from constructions with multiple auxiliaries Van Urk (2018:943), which show that the object occupies a *derived* position. The moved object precedes the verb and any in situ auxiliaries, such as the adverbial auxiliary *dâac* ‘do.quickly’ in (11a–b).

(11) *Object in a derived position before auxiliary:*

- a. Tiik à-cé [<sub>VP</sub> **cuñin** dâac t̩aal].  
 woman D-PRF food do.quickly.NF cook.NF  
 ‘The woman has cooked the food quickly.’
- b. \*Tiik à-cé [<sub>VP</sub> \_\_\_\_ dâac **cuñin** t̩aal].  
 woman D-PRF do.quickly.NF food cook.NF  
 ‘The woman has cooked the food quickly.’

A similar conclusion can be drawn from a causative construction formed with the verb *c̩ɔ̩k* ‘make’ (Van Urk 2015:160). The causative verb embeds a verb phrase and requires promotion of another DP to its object position (12a–b).<sup>2</sup>

(12) *Object in a derived position before causative:*

- a. Tiik à-cé [<sub>VP</sub> **m̩ɔ̩c** c̩ɔ̩k [c̩ɔ̩l m̩ɛ̩t̩]].  
 woman D-PRF man make.NF call child  
 ‘The woman made the man call the child.’
- b. \*Tiik à-cé [<sub>VP</sub> \_\_\_\_ c̩ɔ̩k [c̩ɔ̩l **m̩ɔ̩c** m̩ɛ̩t̩]].  
 woman D-PRF make.NF call man.GEN child  
 ‘The woman made the man call the child.’

These facts follow from an object movement analysis, since the causative introduces its own vP and so hosts an additional object position. The next section demonstrates that this object position provides key evidence for successive-cyclic movement through a clause-internal domain.

## 2.2 Two reflexes of successive cyclic movement

Van Urk (2015, 2018) and Van Urk and Richards (2015) show that the object position described above is also the target for obligatory steps of intermediate movement in long-distance dependencies, as diagnosed by two reflexes of successive cyclicity. First, long-distance movement satisfies the V2 property of every intervening verb phrase, so that any object position on the path of movement must now appear empty. Second, movement of a plural phrase leaves a trail of 3rd person plural pronouns in each object position, which I argue reflects the spell-out of an intermediate copy, following Van Urk (2018).

<sup>2</sup>See Van Urk (2015:sec. 5.4) for more detail on the syntax of the causative construction.

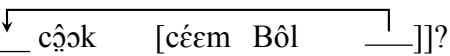
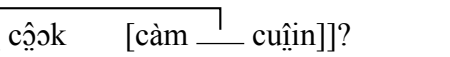
Van Urk (2015) and Van Urk and Richards (2015) demonstrate that long-distance movement satisfies the V2 property of intervening verb phrases, because a clause-medial object position on the path of movement must appear empty. In (13a–d), we see that *wh*-clefting an object in a ditransitive must be preceded by movement to Spec-*vP*, as evident in the impossibility of movement of the other object.

(13) *Object movement must transit through Spec-*vP* in ditransitives:*

- a. Yè **ŋó** cǐi môc [<sub>VP</sub> — yiêen Ayén]?  
 be.3SG what PRF.OV man.GEN give.NF Ayen  
 ‘What has the man given Ayen?’
- b. \*Yè **ŋó** cǐi môc [<sub>VP</sub> **Áyén** yiêen]?  
 be.3SG what PRF.OV man.GEN Ayen give.NF  
 ‘What has the man given Ayen?’
- c. Yè **ŋà** cǐi môc [<sub>VP</sub> — yiêen kítáap]?  
 be.3SG who PRF.OV man.GEN give.NF book  
 ‘Who has the man given the book to?’
- d. \*Yè **ŋà** cǐi môc [<sub>VP</sub> **kítáap** yiêen]?  
 be.3SG who PRF.OV man.GEN book give.NF  
 ‘Who has the man given the book to?’

We see the same effect in extraction from the complement of a causative. Long-distance movement must be preceded by movement to the object position of the causative verb, so that object movement of another nominal is blocked (14a–d).

(14) *Movement must transit through Spec-*vP* in causatives:*

- a. Yè **ŋó** cǐi Áyén [<sub>VP</sub> — còk [céem Ból —]]?   
 be.3SG what PRF.OV Ayen.GEN make.NF eat.OV Bol.GEN  
 ‘What has Ayen made Bol eat?’
- b. \*Yè **ŋó** cǐi Áyén [<sub>VP</sub> **Ból** còk [céem —]]?  
 be.3SG what PRF.OV Ayen.GEN Bol make.NF eat.OV  
 ‘What has Ayen made Bol eat?’
- c. Yè **ŋà** cǐi Ból [<sub>VP</sub> — còk [càm — cuñin]]?   
 be.3SG who PRF.OV Bol.GEN make.NF eat food  
 ‘Who has Bol made eat the food?’
- d. \*Yè **ŋà** cǐi Ból [<sub>VP</sub> **cuñin** còk [càm —]]?  
 be.3SG who PRF.OV Bol.GEN food make.NF eat  
 ‘Who has Bol made eat the food?’

I conclude that long-distance dependencies in Dinka involve intermediate movement steps through the object movement position, Spec-vP.

These facts converge with an unusual effect associated with extraction of plurals. Long-distance movement of a plural phrases triggers *pronoun copying* in each Spec-vP position on the path of movement (Van Urk 2018), triggering the appearance of the 3rd person plural pronoun *ké/kêek*. In (15a), pronoun copying is triggered by object relativization, while the example in (15b) shows successive instances of pronoun copying in a long-distance *wh*- cleft.

(15) *Pronoun copying at each vP edge:*

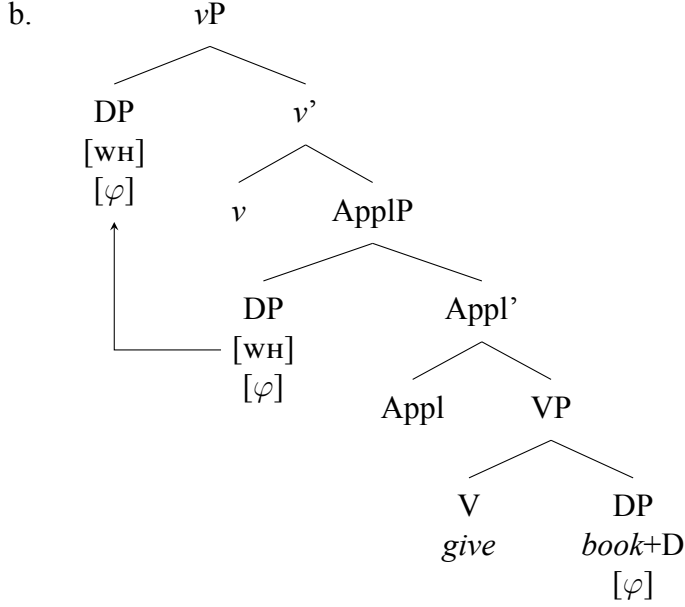
- a. Bòl à-cé **ròoor** [CP cè [vP **ké** lâat]] t̩iŋ.  
 Bol D-PRF men PRF.3SG 3PL insult.NF see.NF  
 ‘Bol has seen the men he has insulted.’
- b. Yè **kôc-kò** y̩i Bòl [vP **kêek** luêeel è c̩i Áyèn [vP **kêek**  
 be.3SG people-which be.OV Bol.GEN 3PL say.NF C PRF.OV Ayen.GEN 3PL  
 t̩iŋ]]?  
 see.NF  
 ‘Which people is Bol saying that Ayen has seen?’

Van Urk (2018) analyzes this effect as multiple spell-out, so that these plural pronouns realize an intermediate copy in the position targeted by object movement.

Van Urk (2015, 2018) and Van Urk and Richards (2015) argue that these obligatory intermediate movement steps can be understood if there is a clause-internal phase in Dinka, which I identify as vP. Because of the Phase Impenetrability Condition, long-distance dependencies can only be successfully formed if the moving phrase undergoes successive-cyclic movement to the edge of each vP on the path of movement. I illustrate with extraction out of a ditransitive, which must obligatorily transit through the object position, as in (16a). The syntactic structure for vP is represented in (16b).

(16) *Derivation of movement out of ditransitive vP:*

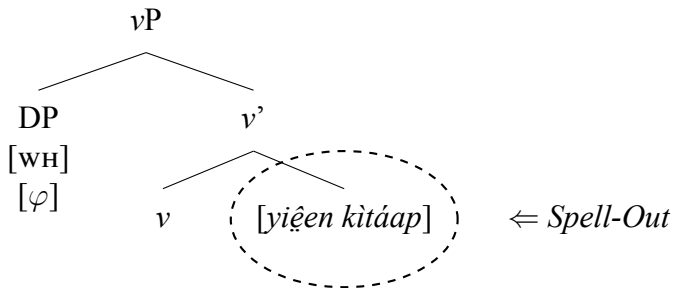
- a. Yè **ŋ̩à** c̩i môc [vP — yîen kitáap]?  
 be.3SG who PRF.OV man.GEN give.NF book  
 ‘Who has the man given the book to?’



I adopt a structure for Dinka ditransitives in which the goal may be introduced in the specifier of an ApplP (see Van Urk 2015:4.1 for more detail). From this position, a *wh*-probe on *v* triggers movement of the goal to the object position in the specifier of *v*P. Recall that, in a syntax-first model, syntactic derivations manipulate only abstract feature bundles (Halle and Marantz 1993).

The phasehood of *v*P results in Spell-Out after *v*P is complete, of all material in the phase except for the phase edge and the phase head itself, *v*. I propose to conceive of Spell-Out as a *replacive* operation, which overwrites the syntactic feature bundle on a head with its PF output after Vocabulary Insertion and application of the phonological grammar (see also Adger 2007; cf. Bobaljik 2000). As a result, the *v*P in (16b) is mapped to the output in (17).

(17) *PF output for (16b) with replacive Spell-Out:*



Any syntactic material in the scope of Spell-Out is converted into a PF representation, so that syntactic features are removed and replaced with phonological content. In the *v*P in (16), the phrase in the complement position of *v*, ApplP, is targeted for Spell-Out, which yields the verb

*yiêen* ‘give.NF’ and the DP *kítáap* ‘book’, once empty structure is eliminated.<sup>3</sup> This replative view explains the Phase Impenetrability Condition, because any material targeted by Spell-Out will no longer be accessible to syntactic operations, which look for syntactic features. As a result, a DP object that does not move to the *v*P edge cannot undergo further movement, so that an example like (13b), repeated in (18), is ungrammatical.<sup>4</sup>

(18) *No movement of direct object out of complement of v:*

\*Yè      **ŋó**      cǐ      môc      [<sub>vP</sub> **Àyén** yiêen            ]?  
 be.3SG what PRF.OV man.GEN      Ayen give.NF  
 ‘What has the man given Ayen?’

In contrast, Spell-Out does not apply to material on the phase edge, such as the goal DP, whose *wh*-feature can continue to be targeted for further movement.

The phasehood of *v*P and the replative nature of Spell-Out then ensure that any long-distance dependency must involve intermediate successive-cyclic movement to the edge of any *v*P on the path of movement, leading to obligatory V2 satisfaction and pronoun copying.

## 2.3 Leapfrogging vs. successive cyclicity

In a recent paper, Keine and Zeijlstra (2025) develop an alternative account of long-distance dependencies in Dinka, which avoids positing a clause-internal phase and makes use of leapfrogging. In this section, I discuss this proposal in more detail and demonstrate that, contra Keine and Zeijlstra, the domain demarcated by the object position and copied *kêek* lies below tense, aspectual morphology, and negation.

Keine and Zeijlstra (2025) agree that Dinka long-distance movement is successive-cyclic and involves intermediate movement steps in the middle field. However, they propose a different rationale for intermediate movement: leapfrogging. Specifically, they posit a restriction in Dinka that all  $\bar{A}$ -dependencies require attraction of the closest nominal. As a result, to extract an object requires moving over any intervening DPs, such as a subject, in addition to object movement. For example, topicalization of an object in their view must also involve a move-

<sup>3</sup>I discuss word formation involving the verb and morphemes like the applicative in more detail in section 4.

<sup>4</sup>In section 4, I propose that the verb is nonetheless able to move for the purposes of V2, via a postsyntactic operation of Raising that happens at PF (Harizanov and Gribanova 2019).

ment step to an intermediate position above the subject. They adopt the idea that subjects move to Spec-TP in Dinka (see Van Urk 2015:ch. 3.2), and posit leapfrogging movement to a second, higher specifier of TP, as schematized in (19b) for the example in (19a).

(19) *Leapfrogging derivation of (19a):*

- a. **Cuḡin** à-céem Áyèn nè pǎal.  
 food 3s-eat.OV Ayen.GEN P knife  
 ‘Food, Ayen is eating with a knife.’
- b.  $[_{CP} \text{Obj } V [_{TP} \text{Obj } [_{T'} \text{Subj} \dots \text{Obj} \dots]]]$

One downside of this account is that OS order is never attested in the middle field, so that these derivations rely on a movement step for which there is no independent evidence. The most significant problems with this alternative, however, arise in the treatment of pronoun copying. In order to tie the idea of leapfrogging to copied *ké/kêek* after the subject, Keine and Zeijlstra treat copied *ké/kêek* as the realization of agreement on a T head, spelling out a reflex of movement to the outer specifier. This probe must be stipulated to only spell out agreement with the second DP moving to one of its specifier, but this proposal allows them to reconcile leapfrogging with the fact that copied *ké/kêek* follows the subject.

The first issue is that, morphologically, *ké/kêek* is a pronoun. As pointed out by Van Urk (2018:sec. 2.2), Dinka pronouns have a weak CV form and a strong CVVC form, and the same alternation obtains in copied *ké/kêek*. A Weak/strong alternation is common in pronouns and provides key motivation for the pronoun copying account developed in Van Urk (2018).

Second, identifying copied *ké/kêek* as T misses key generalizations about the organization of the Dinka clause. Dinka has overt morphology that realizes tense, the past tense prefix *é-*. This prefix always occurs on the clause-level V2 position (20a–b), and so in a higher position than copied *ké/kêek*. As a result, the past tense prefix co-occurs freely with copied *ké/kêek*, in main and embedded clauses (20c).

(20) *Past tense prefix on V2 position:*

- a. Ayén **ée-lèet** wôok.  
 Ayen PST.3s-insult 1PL  
 ‘Ayen was insulting us.’

- b. Wôok **é-kè-cé** tiik yiên kitáap wár akól.  
 1PL PST-PL-PRF woman.GEN give.NF book before day  
 ‘We had given the woman a book yesterday.’
- c. Yè kôoc-kò **é-kè-yá** **ké** tàak [CP **è é-kè-cí** Ayèn  
 be.3SG people-which PST-PL-be.2SG 3PL think.NF C PST-PL-PRF.OV Ayen.GEN  
**ké** gâam gálám]?  
 3PL give.NF pen  
 ‘Which people did you think Ayen had given a pen to?’

Copied *ké/kêek* then is not an exponent of tense.<sup>5</sup> In fact, the past tense prefix *é-* is part of a broader class of functional material that can only occur in the clause-level V2 position. This class also includes the declarative proclitic *à-*, the perfect auxiliary *cé*, the negative proclitic *cé*, and the negative perfect *kêec* (Andersen 2007):<sup>6</sup>

(21) *Clause type, perfect, and negation in second position:*

- a. Ayén **à-cé** [<sub>VP</sub> cuîin câam nè pǎal].  
 Ayen D-PRF food eat.NF P knife  
 ‘Ayen has eaten food with a knife.’ (Bor)
- b. Mariàal **à-cé** bíi dǎaŋ-dè [<sub>VP</sub> mèt tǎat tiik].  
 Marial D-NEG FUT.OV boy-3s child beat.APPL.NF woman  
 ‘Marial’s boy will not beat the child for the woman tomorrow.’
- c. Tiik **à-kêec** [<sub>VP</sub> tǎal].  
 woman D-NEG.PR F COOK.AP.NF  
 ‘The woman has not cooked.’ (Agar; Andersen 1992:7, Andersen 2002:95)

All of these co-occur freely with copied *ké/kêek* and occupy a position above it.

There is functional material that can occur in the *vP* domain, following the object position and thus copied *ké/kêek*.<sup>7</sup> Dinka has a large class of adverbial auxiliaries, called “secondary auxiliary verbs” by Andersen (2007), and these can surface low in the verb phrase after the object position in the presence of a higher auxiliary (22a–b).

<sup>5</sup>See also Keine and Zeijlstra (2025:fn. 18).

<sup>6</sup>There is dialectal variation with the future auxiliary *bé*, which can be in the inner domain in Bor (Van Urk and Richards 2015:p. 123), but remains outside of it in Agar (Andersen 2007:p. 95). This auxiliary is derived from the motion verb *bé* ‘come’ (Andersen 2007:98), and so could variably instantiate a Tense head (Agar) or a semantically bleached motion verb (Bor).

<sup>7</sup>As discussed by Andersen (2007:95–96), the copular verb *yâa* appears in non-finite form in the *vP* domain to mark habitual meaning. This fact could be taken to suggest that the clause-internal domain boundary lies in between perfect and habitual aspect. However, copular verbs can mark aspect via biclausal restructuring constructions (Laka 2006; Coon 2010), and so this structure requires more detailed investigation.

(22) *Adverbial auxiliaries in the vP domain:*

- a. Tiik à-cé [<sub>vP</sub> cuîin **dâac** t̩aaal].  
 woman D-PRF food do.quickly.NF cook.NF  
 ‘The woman has cooked the food quickly.’ (Bor)
- b. ... [bè [<sub>vP</sub> nîn **bêer** l̩ɔk ɲɔɔɲ].  
 FUT.3SG eye do.again.NF do.afterwards.NF be.poor.NF  
 ‘...to be worried again.’ (idiomatic) (Agar; Andersen 2007:102)

These adverbial elements plausibly express meanings generated low in the clause, and so I propose that they realize functional heads inside the clause-internal phase.

Taken together, we arrive at a picture in which copied *ké/kêek* splits the clause into two familiar domains: a lower domain in which argument and event structure is established and a higher domain that hosts TAM morphology (23).

(23) *Organization of the Dinka clause:*

[<sub>CP</sub> Declarative Tense Negation Perfect ... [<sub>vP</sub> *ké/kêek* Adverbials Root]]

As section 3 shows, there is a similar split in the morphology. Argument structure morphology such as the applicative, antipassive, and associated motion marking is generated in the vP phase and can appear after copied *ké/kêek*. But agreement morphology, extraction marking, as well as the passive only occurs in the higher domain, on the clause-level V2 position before *ké/kêek*.

Keine and Zeijlstra’s (2025) proposal that copied *ké/kêek* is the realization of a head that hosts the subject in a leapfrogging derivation is then difficult to reconcile with morphological and syntactic observations about the organization of the clause. On the syntactic side, it is not clear why negation and the perfect obligatorily precede *ké/kêek*.<sup>8</sup> On the morphological side, the observation that passive morphology is always in the higher domain poses a similar problem. I conclude that V2 satisfaction in the verb phrase as well as pronoun copying diagnose a clause-internal domain in Dinka, which I will continue to refer as the vP phase, as it lies below aspect, negation, and passive. It includes the verb, adverbial elements, argument structure morphology, and an object movement position.

<sup>8</sup>It is of course technically possible to propose derivations in which negation, the perfect, and the passive are all generated lower than *ké/kêek*, but have to raise past this position obligatorily, if the Head Movement Constraint is relaxed sufficiently. But such derivations seem to undermine the idea that there is a reliable correspondence between linear order and underlying syntactic structure.

To rescue Keine and Zeijlstra’s (2025) account, the subject position they posit would have to be immediately above this  $\nu$ P domain. If leapfrogging occurs very low, such a proposal becomes difficult to distinguish from the phase-based one, since it posits virtually the same movements, except that it incorporates an additional unmotivated movement step.<sup>9</sup> As a result, this paper turns to an investigation of the morphology and phonology of Dinka, to adjudicate between a phase-based account and leapfrogging. Clear independent evidence for the  $\nu$ P phase emerges in the study of Dinka nonconcatenative morphology. I first introduce the distinction between inner and outer nonconcatenative morphemes, based on (Andersen 1992, 1993 et seq.). I show that this distinction in the Dinka literature lines up with syntactic diagnostics, so that all inner morphology is generated in the  $\nu$ P phase. In the rest of this paper, we will see that the verb plus inner morphemes behaves like a locality domain at PF, as predicted by a theory that incorporates a  $\nu$ P phase.

### 3 Inner and outer nonconcatenative morphology in Dinka

This section demonstrates that the syntactic evidence for a clause-internal domain converges with a division in categories of nonconcatenative morphology on Dinka verbs. A monosyllabic template for verbs means that most morphological categories in Dinka are marked by changes in length, vowel quality, voice quality, and tone. Within these paradigms, Andersen (1992, 1993) distinguishes between outer and inner morphology (“inflectional” and “derivational” morphology, respectively, in Andersen’s terminology). Inner morphology is always expressed on the verb, such as the applicative (APPL), even in clauses with an auxiliary (24), and so appears within the  $\nu$ P phase only. In contrast, outer morphology shifts to the highest auxiliary if one is present and resides in the clause-level 2nd position, as demonstrated for Object Voice morphology (ov) on the future auxiliary in (24).

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<sup>9</sup>One advantage of the leapfrogging account, though, is that it may explain why there is no pronoun copying when the plural subject of an unaccusative undergoes movement. Within the proposal adopted here, pronoun copying is triggered by the probes that establish object shift and intermediate steps of long-distance movement, but not by promotion to subject.

- (24) **Outer morphology (OV)**      **Inner morphology (APPL)**  
 Mariàal à-cè **bǝi**      dǝaŋ-dè      [vp      mèt **tǝat**      tiik ].  
 Marial D-NEG FUT. [OV] boy-3s      child beat. [APPL].NF woman  
 ‘Marial’s boy will not beat the child for the woman tomorrow.’  
 (Agar; Andersen 1992:7)

### 3.1 Monosyllabicity in Dinka roots

This section introduces the key phonological properties that are important for how nonconcatenative morphology works in Dinka. This discussion, and the remainder of the paper, focuses primarily on data from the Agar variety (Andersen 1993, 2014, 2017), as well as Luanyjang (Remijsen and Ladd 2008; Ladd et al. 2009; Remijsen and Manyang 2009). The crucial syntactic and morphological facts are the same for the different varieties (see, for instance, Blum 2023), but the morphophonology of Agar and Luanyjang has been described in more detail.

As in other Western Nilotic languages, roots in Dinka are generally subject to a monosyllabic template. The nouns and verbs from Luanyjang in (25a–h) illustrate.

- | (25) | Noun  | Meaning       | Verb | Meaning |               |
|------|-------|---------------|------|---------|---------------|
| a.   | jòom  | ‘wind’        | e.   | máaan   | ‘hate.NF’     |
| b.   | kít   | ‘color’       | f.   | kóoot   | ‘care.for.NF’ |
| c.   | pjèen | ‘viper’       | g.   | kwǝat   | ‘wrap.NF’     |
| d.   | gwèel | ‘collar bone’ | h.   | tǝet    | ‘pick.NF’     |
- (Luanyjang; Remijsen and Ladd 2008:180,186, Remijsen and Manyang 2009:115,119)

Both the onset and coda consonant are obligatory. Andersen (1993:2) and Remijsen and Manyang (2009:114) give the template in (26), for Agar and Luanyjang, respectively.<sup>10</sup>

- (26) **Surface template for Dinka nouns and verbs:**

C (w) (j) V (V) (V) C

Within the monosyllabic root, a number of contrasts are used to mark morphological categories. First, Dinka vowels display a ternary contrast in length (Remijsen and Gilley 2008):<sup>11</sup>

<sup>10</sup>The *underlying* form of roots is more restricted and maximally permits a long vowel and one glide (Andersen 1993, 2014; Ladd and Blum 2021).

<sup>11</sup>Luanyjang examples are taken from the Remijsen (2013) corpus of noun pairs, where a source is not speci-

| (27) | Short |                   | Long |                 | Overlong |               |
|------|-------|-------------------|------|-----------------|----------|---------------|
|      | láj   | ‘kind of berries’ | làaŋ | ‘kind of berry’ | lăaaŋ    | ‘slave’       |
|      | kít   | ‘color’           | kîit | ‘colors’        | kîiit    | ‘cloth bag’   |
|      | cól   | ‘mouse’           | còol | ‘charcoal’      | còool    | ‘charcoal.PL’ |

Second, most Dinka dialects distinguish three or four tones, high /<sup>ó</sup>/, low /<sup>ò</sup>/, falling /<sup>ô</sup>/, and sometimes also rising /<sup>ǒ</sup>/ . In addition, vowels show a binary contrast between modal/creaky voice (unmarked) and breathy voice /<sup>ᵛ</sup>/, in all vowels except *u*:<sup>12</sup>

| (28) | Unmarked |                   | Breathy |                       |
|------|----------|-------------------|---------|-----------------------|
|      | jáak     | ‘pelican’         | jǎal    | ‘visitor, guest’      |
|      | gwét     | ‘Nile perch’      | gwét    | ‘bead’                |
|      | ŋéɛɛr    | ‘kind of gazelle’ | nwéɛɛr  | ‘Nuer’                |
|      | kîiir    | ‘big river’       | kîiir   | ‘thorny kind of tree’ |
|      | tóoc     | ‘swamp’           | tòon    | ‘pot’                 |
|      | rwòon    | ‘stone of fruit’  | rwòon   | ‘year’                |
|      |          |                   | wúuk    | ‘wing’                |

Finally, alongside these suprasegmental distinctions, changes in vowel quality can mark a morphological category also, through processes of vowel lowering and vowel fronting.

### 3.2 Inner and outer morphology in Dinka

The frequent use of nonconcatenative marking in Dinka makes it hard to tell at first glance how morphemes are ordered. However, it turns out that the domain identified by syntactic diagnostics converges with a morphological distinction between what I call outer (“inflectional”) and inner (“derivational”) morphology. In a study of transitive verbal roots, Andersen (1992) treats every verb as a combination of what he refers to as an “inflectional” category and a “derivational” category. A paradigm of the transitive verb *mîit* ‘pull’ according to this division is provided in Table 2.

cally indicated.

<sup>12</sup>I follow Andersen and mark voice and tone only on the first vowel.

**Table 2. Verbal paradigm of *mîit* ‘pull’.**

|           | Transitive | CF       | CP       | APPL     | APPL.AP  | AP       |
|-----------|------------|----------|----------|----------|----------|----------|
| 1SG       | mjêët      | mjêët    | mjêët    | mjêët    | mjêët    | mjêët    |
| 2SG       | mîit       | mjêët-é  | mîit     | mjêët-é  | mjêët-é  | mjêët-é  |
| 3SG       | mîit       | mîit     | mîit     | mîit     | mîit     | mîit     |
| 1PL       | mîit-kù    | mîit-kù  | mîit-kù  | mîit-kù  | mîit-kù  | mîit-kù  |
| 2PL       | mjêët-kà   | mjêët-kà | mjêët-kà | mjêët-kà | mjêët-kà | mjêët-kà |
| 3PL       | mîit-ké    | mîit-ké  | mîit-ké  | mîit-ké  | mîit-ké  | mîit-ké  |
| PASS      | mîit       | mîit-è   | mîit-è   | mîit-è   | mîit-è   | —        |
| PASS.OBLV | mîit-è     | mîit-è   | mîit-è   | mîit-è   | mîit-è   | mîit-è   |
| SV        | mîit       | mîit     | mîit     | mîit     | mîit     | mîit     |
| OV        | mîit       | mîit     | mîit     | mîit     | mîit     | mîit     |
| NF        | mîit       | mjêët    | mjêët    | mjêët    | mjêët    | mîit     |

(Agar; Andersen 2017:11)

On the horizontal axis of this table, we find different categories of what I will call *inner morphology*, which includes the simple transitive, associated motion marking (centrifugal, CF, and centripetal, CP), the applicative (APPL), and the antipassive (AP). The vertical axis lists categories of what I refer to as *outer morphology*, consisting of agreement, the passive, and Austronesian-style voice morphology (Subject Voice (sv), Object Voice (ov), and Oblique Voice (OBLV)). Andersen includes the non-finite (NF) in this second category as well, though I show below that it patterns syntactically with inner morphology.

I depart from Andersen in not using the labels “inflectional” and “derivational”, because I do not think this distinction reflects an obvious qualitative difference. All of the morphological categories in Table 2 are productive and semantically regular, and none of them change category. In addition, although inner/derivational morphology includes only argument structure morphology, the passive belongs to outer/inflectional morphology.

It is also important to note that, although this division provides a convenient way to visualize paradigms, there is no real sense in which a verb form is necessarily a combination of an inner

and an outer morpheme. First, both Subject Voice (sv) and the simple transitive are not clearly morphologically marked, and so verbs can surface without an inner morpheme and without an outer one. Second, combinations of multiple outer categories (the passive Oblique Voice, PASS.OBLV) and multiple inner categories (the applicative antipassive, APPL.AP) are possible.<sup>13</sup>

Nonetheless, I follow Andersen (1992 et seq.) in taking the distinction between inner and outer morphology to reflect an important generalization about Dinka morphology. Inner morphology is always syntactically lower than outer morphology, as I demonstrate below, because all inner morphology originates in the  $\nu$ P phase identified by syntactic diagnostics. As I argue in section 4, several asymmetries between inner and outer morphology result from the presence of this phase boundary in between the root and outer morphemes.

### 3.3 Inner morphology is in the $\nu$ P phase

This section demonstrates that the  $\nu$ P phase identified in section 2 corresponds to the domain of inner morphology. As noted by Andersen (1992:21), inner morphology is always expressed on the lexical verb, while outer morphology shifts onto the auxiliary in clauses containing auxiliaries. On this basis, I propose that all inner morphology is generated inside the  $\nu$ P phase, while outer morphology is generated outside of it.

In clauses without an auxiliary, the verb raises to second position, to satisfy clause-level V2. All inner and outer morphology is expressed on the verb. The example in (29) demonstrates for the inner category applicative (APPL) and the outer category Object Voice (ov).

(29) *Inner and outer morphology on raised verb:*

| Outer morphology (ov)                                    |         | Inner morphology (APPL) |           |
|----------------------------------------------------------|---------|-------------------------|-----------|
| Wéŋ                                                      | à-mŋit  | ḡḡok                    | [ $\nu$ P |
| cow                                                      | D-pull. | mòc nè ɟwɟɛn            | ]         |
|                                                          | APPL.OV | man P rope              |           |
| ‘The cow, the boy is pulling for the man with the rope.’ |         |                         |           |

(Agar; Andersen 1992:13)

<sup>13</sup>In addition, there are independent explanations for why some combinations are not attested. For example, associated motion marking introduces a location argument and so plausibly realizes the same head as the applicative. Similarly, the passive does not involve promotion to subject and, as a result, no subject agreement occurs. The absence of promotion also explains the lack of a contrast between the Subject Voice and Object Voice in the passive. The remaining inflectional contrasts that are absent can be analyzed as neutralization, such as the loss of a distinction between Subject Voice and Object Voice before subject agreement and the absence of Oblique Voice marking on actives (which seems to be unique to Agar, see Van Urk 2015 on Bor).

The verb originates in the inner  $\nu$ P phase, picks up all inner morphology, and then raises into the left periphery.

Outer and inner morphology diverge when an auxiliary is present, demonstrating that these categories of morphology originate in different domains. Inner morphology stays on the in situ verb, but outer morphology is expressed on the finite auxiliary. The example in (30) illustrates by adding the future auxiliary *bě* to (29). The inner applicative is expressed on the verb, in the  $\nu$ P phase, while outer morphology (Object Voice, ov) is found on the finite auxiliary instead, outside the  $\nu$ P phase.

(30) *Inner morphology on verb, outer morphology on auxiliary:*

| Outer morphology (ov)                                          |            |              |           | Inner morphology (APPL) |             |         |
|----------------------------------------------------------------|------------|--------------|-----------|-------------------------|-------------|---------|
| Mariàal à-cě                                                   | <b>bĭi</b> | ǵáaŋ-dě      | [ $\nu$ P | mět                     | <b>ǵáat</b> | tiik ]. |
| Marial                                                         | D-NEG FUT. | [OV] boy-3SG |           | child beat.             | [APPL].NF   | woman   |
| ‘Marial’s boy will not beat the child for the woman tomorrow.’ |            |              |           |                         |             |         |

(Agar; Andersen 1992:7)

This generalization extends to all inner morphemes, and it follows if all inner morphology is generated in the  $\nu$ P phase.

As evident in (30), in clauses with an auxiliary, non-finite morphology appears on the verb (*ǵáat* ‘beat.APPL.NF’ in (30)). The morphological category non-finite is used in Dinka for any verb or auxiliary not in the clause-initial V2 position, in opposition to the finite verb/auxiliary. In this way, it is akin to a participial form rather than a category that signals the finiteness of the clause as a whole. Andersen (1992 et seq.) lists the non-finite as outer morphology, because the non-finite combines with all inner morphemes and not with other outer categories. But we can readily explain this apparent difference among outer morphemes in a height-based view. I propose that the non-finite instantiates a functional head in the  $\nu$ P phase, and so belongs to the domain of inner morphology. Nothing prevents the non-finite from combining with other inner categories, because nothing rules out such combinations in principle. The non-finite does not combine with outer morphology simply because only finite verbs raise to the clause-level V2 position, as in other well-studied V2 languages.

Combining morphological and syntactic diagnostics, we can further refine our clausal hierarchy for Dinka (31).

- (31) *Clausal hierarchy for Dinka*.<sup>14</sup>
- |                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>CP phase (“outer”)</b>           | <b>vP phase (“inner”)</b>       |
| (C - T - Neg/Perf); Agr/Voice; Pass | v - NF - Appl/CF/CP - AP - Root |

All inner morphology, along with the non-finite, originates in the lower vP phase, and all outer morphology is generated in the CP phase. The result is a hierarchy compatible with broad crosslinguistic generalizations about the clause, with clause type and tense morphology above aspect morphology above argument structure morphology (e.g. Bybee 1985).

The question that arises now is whether the phase boundary posited for syntactic reasons in Van Urk (2015, 2018) and Van Urk and Richards (2015) can be detected in the morphophonology as well. The next section describes two asymmetries in how inner and outer morphology are expressed that reveal that the verb plus inner morphology constitutes a phase at PF also.

## 4 An inner domain at PF

The vP phase boundary gives rise to two asymmetries in how inner and outer morphology is expressed on the verb. First, only inner morphemes show tonal allomorphy based on verb class. Outer morphology marked by grammatical tones contribute a consistent tone. This pattern follows if only the morphosyntactic features of morphemes realized on the same PF cycle can interact for allomorphic relations (Embick 2010; Bobaljik 2012; Moskal 2015). Second, the root plus inner morphemes is the domain over which the monosyllabic restriction holds. As a result, inner morphology is necessarily nonconcatenative, while outer morphology can be marked by suffixation. I argue that this restriction is the result of cyclic prosodification (Guekguezian 2017; Weber 2022), specifically the fact that the inner cycle must be mapped to a wellformed prosodic word, which in Dinka must be a monosyllable.

<sup>14</sup>Slashes indicate heads that do not transparently combine and may occupy the same position. Semi-colons separate heads that either combine or do not co-occur for independent reasons, but for which I have no evidence that bears on order.

## 4.1 Allomorphy based on verb class

The first piece of evidence that the inner domain is a phasal domain comes from the distribution of allomorphy. Most morphological categories in Dinka are associated with changes in the tone in the root. Two patterns of grammatical tone are found: a consistent replacive tone or a tone that varies based on Dinka’s four verb classes. This section demonstrates that the latter pattern is only found in inner morphemes. This generalization follows if allomorphic relations are constrained by cyclic locality (Embick 2010; Bobaljik 2012; Moskal 2015). I derive constraints on allomorphy from replacive Spell-Out: only the features of morphemes spelled out in the same cycle are accessible for allomorphic relations.

Andersen (1993) distinguishes four classes of transitive verbs in Agar Dinka, based on their tonal behavior (see also Blum 2023 on Bor South). Andersen labels these classes according to the underlying length of the root (short or long) and the tone they carry in the transitive non-finite (an inner morpheme in my account). The examples in (32) demonstrate.<sup>15</sup>

(32) *Tone in the non-finite involves four verb classes:*

| CVC/F            | CVC/L           | CVVC/F          | CVVC/H          |
|------------------|-----------------|-----------------|-----------------|
| gûut ‘stab.NF’   | tèeŋ ‘dust.NF’  | mîiit ‘pull.NF’ | téeem ‘cut.NF’  |
| t̪ôor ‘spear.NF’ | nàaj ‘plait.NF’ | lêeer ‘roll.NF’ | ŋáaap ‘open.NF’ |

(Agar; Andersen 1992:23–24,28–29, Andersen 2017:10–13)

Whether any aspect of these tones are underlyingly associated with the root is difficult to establish (see Trommer 2022 and Van Urk 2024 for different views), but what is crucial is the tone associated with the non-finite is not predictable and varies by root. Such patterns of verb class variation in grammatical tone are found with a range of inner morphemes, including the centrifugal, applicative, antipassive, and applicative antipassive (33).

<sup>15</sup>The non-finite also triggers lengthening.

(33) *Verb class variation with inner morphemes:*

| CVC/F                       | CVC/L                       | CVVC/F                                                | CVVC/H                                                           |
|-----------------------------|-----------------------------|-------------------------------------------------------|------------------------------------------------------------------|
| H<br>gúut<br>‘stab.CF’      | F<br>têeŋ<br>‘dust.CF’      | F<br>mîiit<br>‘pull.CF’                               | H<br>téem<br>‘cut.CF’<br><i>Centrifugal</i>                      |
| H<br>gúut<br>‘stab.APPL’    | F<br>têeŋ<br>‘dust.APPL’    | F<br>mîit<br>‘pull.APPL’                              | H<br>téem<br>‘cut.APPL’<br><i>Applicative</i>                    |
| F<br>gûut<br>‘stab.AP’      | F<br>têeŋ<br>‘dust.AP’      | L/H <sup>16</sup><br>mîit/tóook<br>‘pull.AP/crack.AP’ | L<br>têm<br>‘cut.AP’<br><i>Antipassive</i>                       |
| F<br>gûut<br>‘stab.APPL.AP’ | F<br>têeŋ<br>‘dust.APPL.AP’ | F<br>mîiit<br>‘pull.APPL.AP’                          | H<br>téem<br>‘cut.APPL.AP’<br><i>Applicative<br/>antipassive</i> |

(Agar; Andersen 1992:28–29, Andersen 2017:10,12,19)

In contrast, outer morphology conditions patterns of consistent replacive tone. In Agar, Object Voice (ov) is always marked by a replacive high tone, while the passive Oblique Voice is associated with falling tone (34).

(34) *Consistent replacive tone with outer morphemes:*

| CVC/F                           | CVC/L                           | CVVC/F                           | CVVC/H                                                              |
|---------------------------------|---------------------------------|----------------------------------|---------------------------------------------------------------------|
| H<br>gúut<br>‘stab.ov’          | H<br>téeŋ<br>‘dust.ov’          | H<br>mîiit<br>‘pull.ov’          | H<br>téem<br>‘cut.ov’<br><i>Object Voice</i>                        |
| F<br>gûut-è<br>‘stab.PASS.OBLV’ | F<br>têeŋ-è<br>‘dust.PASS.OBLV’ | F<br>mîiit-è<br>‘pull.PASS.OBLV’ | F<br>têeem-è<br>‘cut.PASS.OBLV’<br><i>Passive<br/>Oblique Voice</i> |

(Agar; Andersen 1992:28–29, Andersen 2017:10–13)

<sup>16</sup>CVVC/F verbs in the antipassive have a different paradigm depending on the underlying voice quality of the root (Andersen 2017:sec. 3.2.4).

At first glance, the passive and the 2SG appear to be exceptions to be this generalization, since they are associated with verb class variation in the simple transitive (35).

(35) *Apparent verb class variation with outer passive and 2SG:*

| CVC/F       | CVC/L       | CVVC/F      | CVVC/H     |                |
|-------------|-------------|-------------|------------|----------------|
| H           | H           | F           | H          |                |
| gùt         | tén         | mîit        | téem       | <i>Passive</i> |
| ‘stab.PASS’ | ‘dust.PASS’ | ‘pull.PASS’ | ‘cut.PASS’ |                |
| L           | L           | F           | H          |                |
| gùt         | tén         | mîit        | téem       | <i>2SG</i>     |
| ‘stab.2sg’  | ‘dust.2sg’  | ‘pull.2sg’  | ‘cut.2sg’  |                |

(Agar; Andersen 1992:28–29, Andersen 2017:10,12)

However, these forms are irregular: across paradigms, the passive and 2SG are otherwise marked by a suffix *-e*, a consistent replacive tone, as well as patterns of vowel lowering or fronting and lengthening. All of these effects are absent in (35), suggesting that these forms do not contain a passive or 2SG morpheme at all (see also Van Urk 2024). In contrast, verb class variation is a regular part of the expression of inner morphemes.

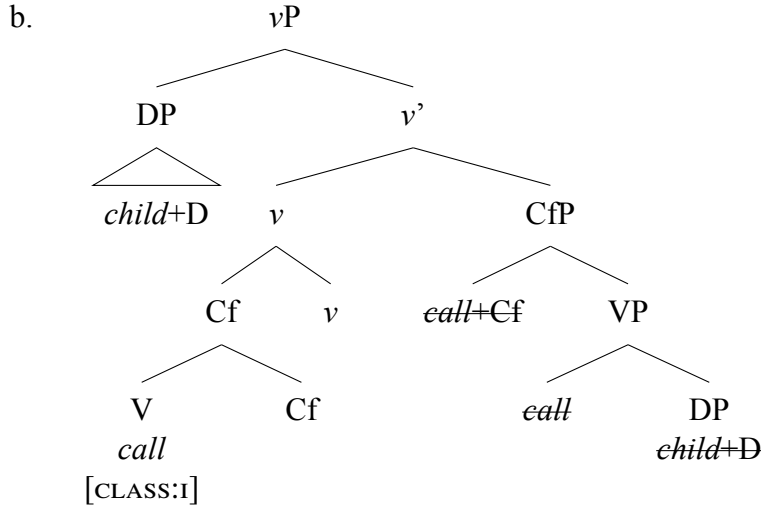
These facts demonstrate that inner and outer morphemes differ in whether they show allomorphy based on the conjugation class of the root. With inner morphemes, verb class variation of this type is commonplace, but it is not part of the regular exponence of outer morphemes. I propose that this generalization follows from the idea that a phase boundary separates inner and outer morphology, adopting the idea that cyclic locality constrains allomorphy relations (Embick 2010; Bobaljik 2012; Moskal 2015; Moskal and Smith 2016).

This locality restriction on allomorphy follows from the idea that the mapping of syntax to PF is replacive in nature. Syntactic features are overwritten by a PF output, so that they can only be visible for allomorphy within a syntactic cycle. I illustrate with the centripetal of *còl* ‘call’ (36a). I propose that Dinka verbs come with a [CLASS] feature, either [CLASS:I] or [CLASS:II]. I put *còl* ‘call’ in Class I, and so it carries the corresponding feature. I identify centrifugal morphology with a head *Cf*. In the first phase, the verb and the centrifugal head combine via head movement and move up to the phase head *v*, while the object moves to Spec-*vP* (36).

(36) *Output of first phase syntax for (36a):*

- a. Tiik à-côôol mèt̚.  
 woman D-call.CF child  
 ‘The woman is calling the child away.’

(Agar; Andersen 2017:21)



To ensure that the verb and inner morphemes form a domain for allomorphic relations, it is important in this account that head movement does not cause the verb to escape the domain of Spell-Out (contra Den Dikken 2007). I propose that the phase head undergoes Spell-Out also (Newell and Piggott 2014). As a result, the phase head *v*, the centrifugal head, and the verb all undergo Spell-Out. The centrifugal morpheme spells out in different ways depending on class feature and length of the verb, as in the insertion rules in (37). With short Class I verbs, the centrifugal is marked by a falling tone.<sup>17</sup>

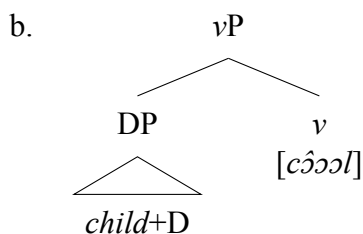
- (37) *call* → *còol*  
 Cf → F / \_\_\_\_ CVC[CLASS:I]

In this view, inner morphemes can display allomorphy for verb class, because they undergo Spell-Out with the verb. The prosodic word consisting of the verb, the centrifugal, and *v* then overwrites the complex head *v* once PF interpretation is complete, so that the class feature is lost, yielding the output in (38b).

<sup>17</sup>The centrifugal is also marked by lengthening and fronting, and at least a floating mora and [+front] feature are likely also part of the realization of the centrifugal (Trommer 2011; Van Urk and Chong 2025).

(38) *Output of first phase PF for (38a):*

- a. Tiik à-côôol mèt.  
 woman D-call.CF child  
 ‘The woman is calling the child away.’ (Agar; Andersen 2017:21)



In order to allow for the verb to undergo further head movement in the syntax, I propose that, alongside the object on the phase edge, the label of the complex head  $v$  is retained, even if Spell-Out is otherwise a replacive operation. The preservation of the features of  $v$  differentiates the phase head from all other material in the Spell-Out domain, but I suggest that the phase head must remain syntactically visible to allow for further instances of External Merge to  $vP$  (see Chomsky 2000:108). An alternative would be adopt a view in which Spell-Out is *partially* replacive, and a category feature is retained that can be targeted to trigger head movement (see also Fenger and Weisser 2025:sec. 4.2.2 and fn. 38 for references suggesting that category information is accessible to phonology).<sup>18</sup>

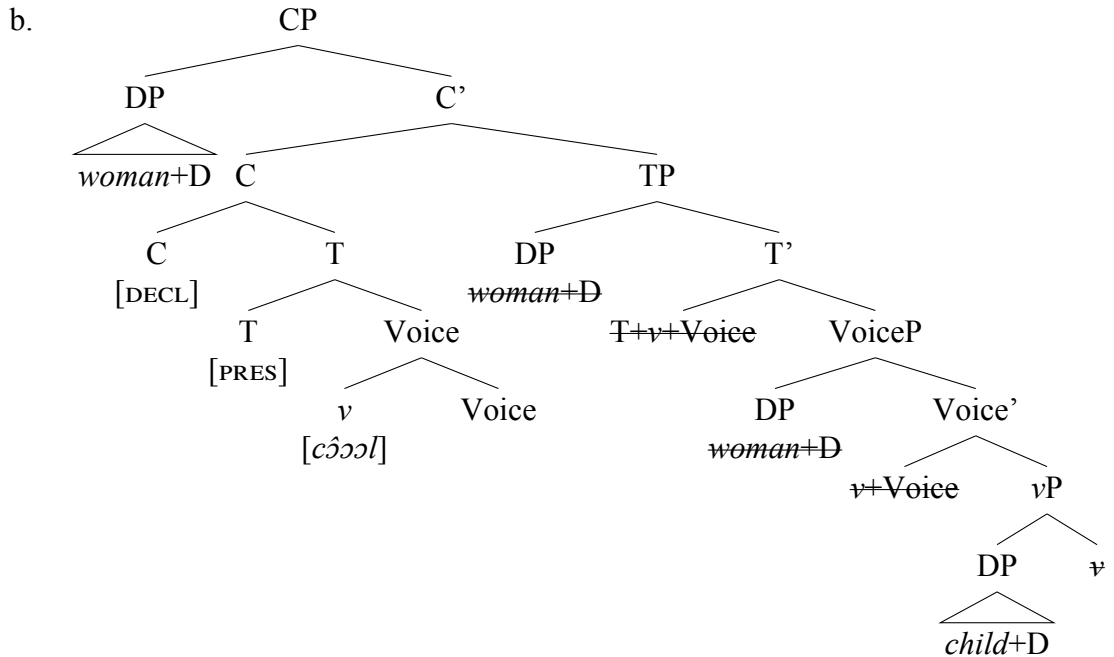
Outer morphemes are introduced in the second phase, which takes as its input the output of PF in (38b). In the CP phase, additional functional structure is introduced, including Voice, which introduces the subject. The head  $v$ , which contains the prosodic word formed by the verb and inner morphemes, moves up to C for the purposes of V2. Because the syntactic label  $v$  is not replaced by Spell-Out, this head can continue to be targeted for syntactic movement. In addition, the subject undergoes movement through Spec-TP to Spec-CP, arriving at the output structure in (39b) for the CP domain.

<sup>18</sup>Another explanation for why head movement does not alter the scope of Spell-Out that is compatible with my proposal would be to adopt the idea that at least some instances of head movement reflect PF operations (Chomsky 2001; Platzack 2013; Harizanov and Gribanova 2019). In Dinka, C and T are typically realized as left-peripheral proclitics. Movement of the highest verb or auxiliary to the left periphery could in principle be analyzed as attraction of the closest prosodic word to serve as the host for these proclitics. If so, movement for V2 would take place after Spell-Out.

(39) *Output of second phase syntax for (39a):*

- a. Tiik à-côôol mèt.  
 woman D-call.CF child  
 ‘The woman is calling the child away.’

(Agar; Andersen 2017:21)



Outer morphemes are inserted upon Spell-Out of this phase. In this second cycle of PF interpretation, the class features associated with the verb are no longer visible, so that allomorphy for conjugation class is not longer possible. In accordance with this prediction, TAM morphemes are invariant across verb classes and show no allomorphy effects based on the identity of the verb. In line with this view, TAM morphemes do affect each other for the purposes of allomorphy. The 3PL suffix *-kè*, for example, spells out number agreement on the declarative proclitic *à* only in the past tense (Andersen 2014:225).

In this model of Spell-Out, what is visible to outer morphemes is only the phonological content of the verb. As a result, outer morphemes can only show allomorphy based on *phonological* features. There is a set of outer morphemes, the plural suffixes, that conditions a pattern of tonal allomorphy, but based on the length of the verb alone and not verb class, in line with this prediction. In the simple transitive, for example, short roots carry a high tone before a plural suffix, and long roots a low tone, as in (40).

(40) *Length-based variation with plural suffixes:*

| CVC/F                          | CVC/L                          | CVVC/F                          | CVVC/H                         |
|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| H                              | H                              | L                               | L                              |
| gút-kù ‘stab-1 <sub>PL</sub> ’ | tén-kù ‘dust-1 <sub>PL</sub> ’ | mìit-kù ‘pull-1 <sub>PL</sub> ’ | tèem-kù ‘cut-1 <sub>PL</sub> ’ |
| gút-kè ‘stab-3 <sub>PL</sub> ’ | tén-kè ‘dust-3 <sub>PL</sub> ’ | mìit-kè ‘pull-3 <sub>PL</sub> ’ | tèem-kè ‘cut-3 <sub>PL</sub> ’ |

(Agar; Andersen 1992:28–29, Andersen 2017:10,12)

This length-based allomorphy pattern is possible given the input to the second cycle of Spell-Out in (39), since the phonological content of the verb remains visible. All other outer categories display only patterns of regular tone across verbs and so also do not show verb class variation.<sup>19</sup>

A key question for a theory in which phases impose restrictions on allomorphy is how to understand crosslinguistic variation in the accessibility of functional morphemes in the CP phase. Much previous work that uses phase theory to account for morphophonological domains has pointed out the existence of languages like Dinka, in which complex words contain a domain boundary that appears to align with a verb phrase (Newell 2008; Fenger 2020; Guekguezian 2021; Fenger and Weisser 2025). However, at the same time, it is well-known that there are languages in which TAM morphology can be local to the verb, as in the well-known patterns of tense-based allomorphy in English. I will not take up the task of developing here a full account of verbal allomorphy across languages, but I conclude from this variation that PF interpretation after Spell-Out need not be successful. In particular, I propose that a complex head *v* containing the verb may sometimes fail to be replaced with phonological content if PF-specific wellformedness conditions prevent successful phonologization. Bobaljik and Wurmbrand (2013) propose, for example, that the existence of an insertion rule that references the first head outside of the Spell-Out domain can suspend the application of Spell-Out. Similarly, Fenger and Weisser (2025:sec. 4.2.2) argue, based on complex patterns of domain formation in Sinhala umlaut, that Sinhala maintains a PF restriction that prevents verb roots from being phonologized in isolation, so that the first affix is always local to the verb. I suggest then that such PF conditions can delay replacive Spell-Out of the phase head *v* to a second cycle, relaxing the locality conditions on allomorphy in some languages (see also Kilbourn-Ceron et al. 2016 for discussion).

<sup>19</sup>There is some debate about whether other outer categories contribute tone at all (see Trommer 2022 and Van Urk 2024 for discussion).

## 4.2 Inner morphemes obey the monosyllabic template

In this section, I discuss another asymmetry between outer and inner morphemes that provides evidence for a clause-internal domain. The root plus inner morphemes is the constituent over which the monosyllabic template is enforced. All inner morphology is expressed via changes to the root, and only outer morphemes can be expressed via affixation. I adopt an approach in which apparently templatic effects within complex words arise because of cyclic formation of prosodic structure (Guekguezian 2017; Weber 2022). The inner cycle must be mapped to a wellformed prosodic word, which in Dinka is a monosyllable.

As first noted by Andersen (1992:9), inner morphology is always marked by nonconcatenative changes in the root. To see this generalization, consider the distribution of outer and inner morphology in the verbal paradigms, illustrated again with the verb *mîit* ‘pull’ in Table 3.

**Table 3. Suffixation in the verbal paradigms of *mîit* ‘pull’.**

|           | Transitive       | CF               | CP               | APPL             | APPL.AP          | AP               |
|-----------|------------------|------------------|------------------|------------------|------------------|------------------|
| 1SG       | mjêet            | mjêet            | mjêet            | mjêet            | mjêet            | mjêet            |
| 2SG       | mîit             | mjêet- <b>é</b>  | mîit             | mjêet- <b>é</b>  | mjêet- <b>é</b>  | mjêet- <b>é</b>  |
| 3SG       | mîit             | mîit             | mîit             | mîit             | mîit             | mîit             |
| 1PL       | mîit- <b>kù</b>  | mîit- <b>kù</b>  | mîit- <b>kù</b>  | mîit- <b>kù</b>  | mîit- <b>kù</b>  | mîit- <b>kù</b>  |
| 2PL       | mjêet- <b>kà</b> | mjêet- <b>kà</b> | mjêet- <b>kà</b> | mjêet- <b>kà</b> | mjêet- <b>kà</b> | mjêet- <b>kà</b> |
| 3PL       | mîit- <b>ké</b>  | mîit- <b>ké</b>  | mîit- <b>ké</b>  | mîit- <b>ké</b>  | mîit- <b>ké</b>  | mîit- <b>ké</b>  |
| PASS      | mîit             | mîit- <b>è</b>   | mîit- <b>è</b>   | mîit- <b>è</b>   | mîit- <b>è</b>   | —                |
| PASS.OBLV | mîit- <b>è</b>   | mîit- <b>è</b>   | mîit- <b>è</b>   | mîit- <b>è</b>   | mîit- <b>è</b>   | mîit- <b>è</b>   |
| SV        | mîit             | mîit             | mîit             | mîit             | mîit             | mîit             |
| OV        | mîit             | mîit             | mîit             | mîit             | mîit             | mîit             |
| NF        | mîit             | mjêet            | mjêet            | mjêet            | mjêet            | mîit             |

(Agar; Andersen 2017:12)

Table 3 displays a mix of nonconcatenative morphology and suffixation. But all suffixes (bolded) express outer morphology only.<sup>20</sup> In contrast, inner morphology, including the non-finite, is

<sup>20</sup>The 2SG and passive suffixes -*é* undergo an independent process of deletion in the simple transitive (the 2SG is also deleted in the centripetal).

never marked by affixation, and is only associated with nonconcatenative processes, such as changes in vowel quality, voice quality, length, and tone. This observation extends to the limited inventory of prefixes/proclitics, such as the declarative particle and past tense *é-* (41).

(41) *Outer prefixes:*

Declarative particle    *à/áa-*

Past                            *é-*

These prefixes combine with all finite forms in Table 3 and are outer morphemes, since they express functional heads in the outer phase. A similar asymmetry appears in related Western Nilotic languages, such as Anywa (Reh 1996) and Nuer (Reid 2019). In these languages too, all inner morphemes are expressed via nonconcatenative morphology, while outer morphemes can be realized via affixation.

I propose to understand the existence of apparently templatic effects within subparts of complex words as a result of cyclic prosodification (Guekguezian 2017, 2021; Weber 2022). In particular, I adopt the idea that, as part of the PF evaluation of phases in a cyclic interleaving model, prosodic structure is built. The mapping from syntactic structure to prosodic structure is regulated by Match-theoretic constraints in the sense of Selkirk (2009, 2011), which favor isomorphism of prosodic and syntactic structure. The constraint **MATCHWORD**, given in (42), dictates that a morphological word, taken here to be a complex head formed by morphosyntactic operations, is mapped to a prosodic word.

(42) **MATCHWORD** (based on Elfner 2012:26–28):

For a morphological word *W* that exhaustively dominates a set of one or more terminal nodes *M*, assign a violation mark if there is no prosodic word ( $\omega$ ) in the phonological representation that exhaustively dominates all and only the phonological exponents of the terminal nodes in *M*.

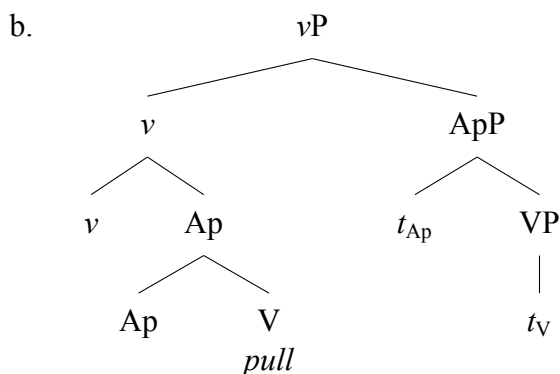
Within a cyclic interleaving model, morphological words formed within the first phase are special in that **MATCHWORD** will force them to map to a well-formed prosodic word, even if later affixation will occur. Guekguezian (2017, 2021) argues that this situation can lead to the formation of recursive prosodic words in complex words that span multiple phases.

I argue that, in Dinka, a syllable functions as the optimal prosodic word. This monosyllabic preference is evident across the language. Adjectives and nouns are always monosyllabic and never carry affixes,<sup>21</sup> and almost all of the functional inventory is monosyllabic also.<sup>22</sup>

I demonstrate how this Match-theoretic calculus explains the absence of affixation in the  $\nu$ P phase, adopting the replacive view of Spell-Out articulated in section 2.2 and 4.1. Consider the example in (43), the verb *m̩it* ‘pull’ marked with an inner antipassive morpheme. Within the  $\nu$ P, the verb and antipassive head *Ap* move up to the phase head  $\nu$ .

(43) *Output of first phase syntax for (43a):*

- a. Ḑḑok à-m̩it.  
 boy D-pull.AP  
 ‘The boy is pulling.’ (Agar; Andersen 2017:4)



Spell-Out is triggered upon completion of the first phase, targeting the complex head  $\nu$  including the verb and antipassive. This head is obligatorily mapped to a monosyllabic prosodic word, following a Match-theoretic calculus based on **MATCHWORD**. This phonological structure over-

<sup>21</sup>Nouns can host demonstratives and possessors, but, as Andersen (2020:293,298) shows, these elements are phrasal enclitics.

<sup>22</sup>Concretely, I propose that Dinka and related Western Nilotic languages prefer for words to correspond to a heavy unary foot. This monosyllabic preference could be modeled by the combination of constraints forcing feet to take this shape and constraints penalizing unparsed syllables as well as more than one foot in a word (see de Lacy 2003; Ketner 2006; Ito and Mester 2015; Uchihara and Mendoza Ruiz 2021). See also Ito and Mester (2015), who argue, on the basis of the distribution of Danish *stød*, that languages can prefer a monosyllabic foot that is co-extensive with the prosodic word.

writes the syntactic features associated with this complex head, retaining the label *v* (44b).

(44) *Output of first phase PF interpretation for (44b):*

- a. D̥ɔk à-m̥it.  
 boy D-pull.AP  
 ‘The boy is pulling.’ (Agar; Andersen 2017:4)
- b.  $\nu$ P  
 |  
 $\nu$   
 [<sub>ω</sub> m̥it]

Cyclic interpretation then ensures that the root plus inner morphemes must correspond to a monosyllabic word.

This proposal explains the observation that, across Western Nilotic languages, argument structure morphology is typically restricted to the monosyllabic root. When outer morphemes undergo PF interpretation, these languages appear to integrate them in two different ways, recursive word formation and prosodic integration, as schematized in (45) and (46). Both options have been argued for as ways of incorporating stray functional morphemes into larger prosodic constituents (see, for instance, Selkirk 2011, Ito and Mester 2012, and Bennett 2018 on prosodic word recursion and Selkirk 1996, Peperkamp 1997, and Gordon and Applebaum 2010 on the fusion of prosodic constituents). In Anywa and Nuer, all outer morphemes appear to be expressed via affixation (Reh 1996; Reid 2019). I propose that these languages always form complex recursive prosodic words, in which a second prosodic word is formed that corresponds to the verb plus inner and outer morphemes, within which the inner monosyllabic prosodic word is contained. In Dinka, such a structure is formed when a verb with its inner morphemes (such as the antipassive) combines with an outer suffix like the 1<sub>PL</sub> *-kù* or the declarative proclitic *à-* (45).

(45) *Recursive word formation:*

- [<sub>ω</sub> [<sub>ω</sub> m̥it]-kù]  
 pull.AP-1<sub>PL</sub>
- [<sub>ω</sub> à- [<sub>ω</sub> m̥it]]  
 DECL- pull.AP

(46) *Prosodic integration:*

- [<sub>ω</sub> mj̥eet]  
 eat.AP.1<sub>SG</sub>

A second option is to integrate outer morphemes into the inner prosodic word (46). This strategy is found in Dinka and Shilluk (see Remijsen and Ayoker 2022), in which outer morphemes can be expressed via nonconcatenative morphology. The 1sg in Dinka, for instance, is expressed via vowel lowering and lengthening and so is integrated into the inner prosodic word.<sup>23</sup> This option is compatible with the view of Spell-Out I develop, since the phonological content of the verb remains accessible, as I argue in more detail in section 5.

In this way, the observation that Western Nilotic languages maintain a strict monosyllabic template that restricts the expression of inner morphemes can be understood as the effect of cyclic spell-out. This template represents an interpretation procedure that applies to an intermediate syntactic structure and maps the morphological word headed by the verb to a wellformed prosodic word. Such effects then provide evidence from prosodic structure for a clause-internal cycle that lines up with the syntactic domain identified in section 2.

## 5 No phase impenetrability in phonology

A key question in work on cyclic locality in morphology and phonology is the extent to which cycles of Spell-Out constrain the application of phonological rules (Newell 2008; Embick 2013). Some phonological changes in Dinka are indeed limited to the first phase. Only inner morphemes or outer morphemes on auxiliaries can introduce a change to breathy voice. However, otherwise, both inner and outer morphology trigger the same nonconcatenative processes of vowel lowering and vowel fronting, altering the vowel quality, length, and tone of the root in comparable ways. As a result, Dinka nonconcatenative morphology provides evidence that there is *no* strict inalterability in phonological cycles. This conclusion follows from a replacive view of Spell-Out, since the output of PF interpretation remains accessible to outer cycles. To explain why some phonological features must be preserved from one phase to the next, I adopt phase faithfulness constraints in the sense of Šurkalović (2013) and McPherson (2014).

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<sup>23</sup>Technically, I propose that there are constraints in the phonological grammar that enforce the preservation of the prosodic word established in the first phase, which I call IDENT<sub>PHASE</sub>( $\omega$ ) (see section 5 for more detail on phase faithfulness constraints). To distinguish between languages like Anywa/Nuer and Dinka/Shilluk, different versions of such a constraint could be used, one that prohibits inserting any new material into an existing prosodic word, high-ranked in Anywa/Nuer, and one that merely prohibits any new prosodic structure, such as a syllable, allowing for nonconcatenative marking.

## 5.1 Opacity in voice quality

I first demonstrate that there are opacity effects in the phonological grammar that align with the clause-internal domain boundary. Since inner and outer morphology are marked by the same nonconcatenative processes, as I discuss in more detail in the next section, it is possible to directly compare their ability to change properties of the root. An asymmetry arises in vowel fronting that reveals that the inner phase is a separate cycle of evaluation. Only inner morphemes that trigger fronting, and not outer ones, can introduce a shift to breathy voice for underlyingly creaky roots, so that voice quality becomes *fixed* in the inner domain.

Many inner categories are marked by vowel fronting. Among these, vowel fronting is often associated with a shift of the root vowel to breathy voice. Recall from section 3.1 that Dinka contrasts two voice qualities on vowels, creaky or modal voice and breathy voice. In the inner categories applicative (47), centripetal (48), and antipassive (49), underlyingly creaky roots are shifted to a breathy voice quality.

(47) *Vowel fronting adds breathy voice in applicative:*

| Root                       | Applicative | Meaning      |
|----------------------------|-------------|--------------|
| (fronting + breathy voice) |             |              |
| bàr                        | bḗ̃ɛr       | ‘take along’ |
| nàj                        | nḗ̃ɛj       | ‘plait’      |
| ḡàaḡ                       | ḡḗ̃ɛḡ       | ‘open’       |

(Agar; Andersen 1992:29, Andersen 2017:11,14)

(48) *Vowel fronting adds breathy voice in centripetal:*

| Root                       | Centripetal | Meaning      |
|----------------------------|-------------|--------------|
| (fronting + breathy voice) |             |              |
| bàr                        | bḗ̃ɛr       | ‘take along’ |
| tèem                       | tḗ̃eem      | ‘cut’        |
| tòoc                       | tḗ̃uuc      | ‘send’       |

(Agar; Andersen 1992:29,37, Andersen 2017:11)

(49) *Vowel fronting adds breathy voice in antipassive:*

| <b>Root</b> | <b>Antipassive</b>         | <b>Meaning</b> |
|-------------|----------------------------|----------------|
|             | (fronting + breathy voice) |                |
| bàr         | bê̤ər                      | ‘take along’   |
| màan        | mê̤ən                      | ‘hate’         |
| ràak        | rò̤k                       | ‘milk’         |

(Agar; Andersen 1992:41,42, Andersen 2017:11,18)

Since this shift to breathy voice can result in raising of the vowel as well, as evident in the centripetal form *tùuuc* ‘send’ in (48) (see Andersen 2017:sec. 4 for more detail), these examples also showcase a process of vowel raising that is limited to the inner domain.

Outer morphology can be expounded by vowel fronting also, in the 3SG, the Oblique Voice passive, and Object Voice. But vowel fronting across these categories never changes the voice quality of the root vowel. Underlyingly creaky vowels remain creaky and breathy vowels remain breathy (50). (Recall that only roots with *a/a* show overt evidence of fronting.)

(50) *Vowel fronting never changes voice quality in outer morphology:*

| <b>Root</b> | <b>3SG</b> | <b>Oblique Voice passive</b> | <b>Object Voice</b> | <b>Meaning</b> |
|-------------|------------|------------------------------|---------------------|----------------|
|             | (fronting) | (fronting)                   | (fronting)          |                |
| bàr         | bè̤ər      | bê̤ər-è̤                     | bé̤ər               | ‘take along’   |
| miit        | miit       | mî̤it-è̤                     | mí̤it               | ‘pull’         |
| tà̤r        | tè̤ər      | tê̤ər-è̤                     | té̤ər               | ‘spear’        |
| gṳ̀t        | gṳ̀t       | gṳ̂t-è̤                      | gṳ́t                | ‘stab’         |

(Agar; Andersen 2017:10–13)

The voice quality of the root then can only be changed in the inner domain.

Independent evidence that voice quality is established in the first phase comes from the paradigm of the perfect auxiliary *cè̤*. This auxiliary only appears in the outer domain of the clause and so does not combine with any inner morphemes. But the perfect auxiliary does combine with outer morphemes that trigger vowel fronting, and, in these combinations, vowel fronting with an outer morpheme *does* introduce breathy voice. As Table 4 shows, whether the perfect auxiliary is creaky or breathy depends on whether it combines with an outer morpheme

that triggers fronting or lowering on verbs.<sup>24</sup>

**Table 4. Partial paradigm of perfect auxiliary *cɛ́*.**

| Process in verbs: |                    |          |
|-------------------|--------------------|----------|
| SV                | <i>cɛ́</i>         |          |
| OV                | <b><i>cɛ́i</i></b> | fronting |
| 1SG               | <i>cà</i>          | lowering |
| 2SG               | <i>cá</i>          | lowering |
| 3SG               | <b><i>cɛ̀</i></b>  | fronting |
| 2PL               | <i>câak</i>        | lowering |
| PASS              | <b><i>cɛ́i</i></b> | fronting |

(Agar; Andersen 2007:93)

These facts follow if voice quality is fixed upon Spell-Out. For verbs, the first phase is the inner domain *vP*, and so only inner morphemes trigger breathy voice. However, for auxiliaries generated in the outer domain, the first phase is the CP, and so outer morphemes can affect voice quality also.

The distribution of voice quality then offers evidence that the inner domain is a phase in the phonology and suggests that at least altering voice quality must occur within a phase. However, I show in the next section that other phonological properties, such as vowel quality, length, and tone, can be altered across the inner domain boundary.

## 5.2 The absence of opacity in vowel fronting and lowering

Nonconcatenative morphology in Dinka is typically expressed through two processes of non-concatenative marking: vowel fronting and vowel lowering. Inner and outer morphemes make use of these processes with equal freedom. These effects offer evidence then that there is no strict phase impenetrability in phonology.

<sup>24</sup>It may seem surprising that the default form of the auxiliary *cɛ́* appears to be breathy, because verbs never display a shift from underlying breathy voice to creaky. However, the short *ɛ* appears to be a default vowel, and is argued to be underlyingly featureless in Van Urk and Chong (2025). This idea would explain why, with this auxiliary, voice quality is determined entirely by the affix.

Andersen (1993) describes nonconcatenative marking in Dinka via three distinct vowel “grades”, laid out for Agar vowels without underlying onset glides in Table 5.

**Table 5. Agar Dinka vowel grades.**

|                             | <b>Creaky</b> |   |   |   |    | <b>Breathy</b> |    |    |    |    |     |
|-----------------------------|---------------|---|---|---|----|----------------|----|----|----|----|-----|
| <b>Root vowel (Grade 1)</b> | i             | e | a | ɔ | o  | ĩ              | ẽ  | ǎ  | ɔ̃ | õ  | u   |
| <b>Fronting (Grade 2)</b>   | i             | e | ɛ | ɔ | o  | ĩ              | ẽ  | ɛ̃ | ɔ̃ | õ  | u   |
| <b>Lowering (Grade 3)</b>   | jɛ            | ɛ | a | a | wɔ | jɛ̃            | ɛ̃ | ǎ̃ | ǎ̃ | ɔ̃ | wɔ̃ |

Each row lists all vowels found in a grade, and each column indicates how vowels correspond across grades. Grade 1 indicates the underlying vowel in the root.<sup>25</sup> Grade 2 describes the vowel that results from vowel fronting. Both inner and outer categories may be characterized by this fronting process, as already noted in the previous section. The 3SG and the applicative are examples of an outer and inner category, respectively (51).

(51)

| <b>Root</b> | <b>3SG</b> | <b>Applicative</b> | <b>Meaning</b> |
|-------------|------------|--------------------|----------------|
|             | (fronting) | (fronting)         |                |
| t̩àr        | t̩ɛ̃r      | t̩ɛ̃r              | ‘spear’        |
| g̩ùt        | g̩ùt       | g̩ùt               | ‘stab’         |
| m̩iit       | m̩iit      | m̩ĩt              | ‘pull’         |

(Agar; Andersen 2017:10–12)

The low central vowel *a* fronts, while other vowels remain unchanged. This change occurs both in inner and outer morphology, and is not prevented from applying in outer morphology.<sup>26</sup>

Grade 3 refers to a process of vowel lowering. All vowels except *a* lower and peripheral vowels also undergo breaking to a diphthong. The examples in (52) illustrate for an outer category (1SG) and an inner one (the non-finite).

<sup>25</sup>This vowel is most frequent and surfaces in unmarked forms (see Andersen 1993, 2017).

<sup>26</sup>One difference in application is that vowel fronting has a greater range in the inner domain than in the outer domain. Andersen (2017:sec. 3.3) points out that vowel fronting optionally fronts underlying *ɔ* to *ɛ̃*, but only as a result of fronting triggered by inner categories. This difference can also be explained through phase faithfulness constraints.

|      |                             |            |                                    |                |
|------|-----------------------------|------------|------------------------------------|----------------|
| (52) | <b>Root</b>                 | <b>1SG</b> | <b>NF Applicative<sup>27</sup></b> | <b>Meaning</b> |
|      |                             | (lowering) | (lowering)                         |                |
|      | tàr                         | tàar       | táar                               | ‘spear’        |
|      | gùt                         | gwòot      | gwóot                              | ‘stab’         |
|      | mìit                        | mjàæt      | mjàet                              | ‘pull’         |
|      | (Agar; Andersen 2017:10–12) |            |                                    |                |

As with fronting, vowel lowering is not restricted to inner categories, and outer categories can alter vowel quality in the same fashion. Such facts offer evidence that phonological structure remains accessible on later cycles.

These observations hold not only of changes in vowel quality, but also of changes in vowel length and tone. Vowel fronting and vowel lowering are reliably associated with lengthening across all morphological categories, of short roots to long and long roots to overlong. In addition, a change in grade typically goes hand in hand with a change in tone also. Such changes in tone are equally possible with inner and outer categories.

Dinka nonconcatenative morphology then provides evidence that there is no strict phonological phase impenetrability (cf. Embick 2013). Although phase boundary separates inner and outer morphology, the inner domain is not inalterable. Higher affixes can induce changes in vowel quality, length, and grammatical tone, just as lower affixes can. I conclude that, although some phonological properties *can* become fixed within a cyclic domain, the output of PF remains fully accessible to later applications of the phonological grammar.

I propose that the absence of phase impenetrability for the phonological grammar is a natural consequence of how cyclic interpretation operates. I argued for a conception of Spell-Out as a replacive operation, that converts a hierarchical syntactic structure annotated with syntactic features into a wellformed phonological representation. As a result of this mapping, intermediate syntactic structure is discarded, giving rise to the Phase Impenetrability Condition. The next syntactic cycle takes the output of PF and adds syntactic structure. But the output of PF is the end-product of the mapping from syntax to PF, and so this output is never discarded. As a con-

<sup>27</sup>Most inner categories are associated with vowel fronting, but the non-finite is expressed via vowel lowering across paradigms. The only exceptions the transitive non-finite, which is irregularly Grade 1, so I illustrate with the non-finite applicative.

sequence, phonological structure remains accessible in later cycles, and phase impenetrability is a property only of rules that operate on syntactic features, such as movement and allomorphy.

Although there is no strict phase impenetrability for phonological rules in this model, we have seen that opacity effects based on phase boundaries do arise. I propose that such opacity effects arise from a phasal architecture in a different fashion. I adopt the idea that the phonological grammar may include phase faithfulness constraints, in the sense of Šurkalović (2013), McPherson (2014), and McPherson and Heath (2016). A schema for this kind of constraint is given in (53).

- (53) IDENT<sub>PHASE</sub>[X]: Assign a violation mark for a feature X present in a phase in the input that is not present in the output.

This constraint represents a pressure to preserve phonological qualities associated with a cycle. High ranking of an IDENT<sub>PHASE</sub> constraint within an Optimality-theoretic grammar can have the result that phonological properties become fixed via phasal evaluation, without any necessary difference between those phonological properties that must be inherited in later computation and those that do not.

For Dinka, high ranking of a IDENT<sub>PHASE</sub>[±breathy] constraint ensures that voice quality is set in the inner phase. The tableaux in (54) and (55) demonstrate for the inner category centripetal and outer category 3SG on the creaky root *bàr* ‘take along’ (Andersen 2017:13). Both trigger vowel fronting, which I attribute to a moraic suffix associated with the features [+front] and [+breathy] (Trommer 2015, Van Urk and Chong 2025).

- (54) *Centripetal triggers vowel fronting and shift to breathy:*

| <b>Input:</b> bàr+μ <sub>[+front,+breathy]</sub> | IDENT <sub>PHASE</sub> [±breathy] | MAX[+breathy] |
|--------------------------------------------------|-----------------------------------|---------------|
| ☞ a. bɛ̃ɛr                                       |                                   |               |
| b. bɛ̃ɛr                                         |                                   | *!            |

(55) *3SG triggers vowel fronting only:*

| <b>1st phase output:</b> bàr                                                               |                                   |               |
|--------------------------------------------------------------------------------------------|-----------------------------------|---------------|
| <b>Input:</b> bàr+μ <sub>[+front,+breathy]</sub>                                           | IDENT <sub>PHASE</sub> [±breathy] | MAX[+breathy] |
| a. bɛ̃ɛr                                                                                   | *!                                |               |
|  b. bɛ̃ɛr |                                   | *             |

In the PF evaluation of the inner phase (56), the centripetal introduces a shift to breathy voice, because the high ranking of a faithfulness constraint favoring breathiness, MAX[+breathy]. But, in the second phase, faithfulness to the output to the first phase ensures that creaky voice is preserved. At the same time, changes in vowel quality, length, and tone can still occur in the outer cycle, because faithfulness constraints for those are low-ranked.

Dinka offers evidence that there is no qualitative difference between those phonological properties that can be altered in later cycles of computation and those that cannot. Vowel quality and voice quality are both properties of the vowel nucleus, yet differ in whether they are alterable. If opacity is enforced by the language-specific ranking of particular phase faithfulness constraints, we correctly predict that no qualitative difference between such properties needs to obtain. The general consequence of this model is only that, even though no aspect of the output of previous cycles is inaccessible, morphemes in inner domains will generally be able to effect greater changes (see also Newell 2008).

## Conclusion

This paper has argued that cyclicity in Dinka offers evidence for one conception of a cycle across modules. Syntactic diagnostics for a clause-internal domain line up with a morphological distinction between inner and outer morphology, and the same phasal domains constrain syntactic, morphological, and phonological computation. These facts provide novel evidence for a clause-internal phase (in line with Chomsky 2001; Rackowski and Richards 2005; Van Urk and Richards 2015, but contra Keine and Zeijlstra 2025). In addition, the comparison of syntactic, morphological, and phonological opacity provides evidence against the idea that an accessibility condition like phase impenetrability constrains the phonological grammar, a gen-

eralization that I derived from a replacive view of Spell-Out. In this view, the similarities and differences in how cyclicity operates across modules follow from the architecture of cyclic interpretation.

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